# Locking in TC

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# Agenda

- Current locking
- RCU completeness
- Lockless qdisc
- TX queue mapping

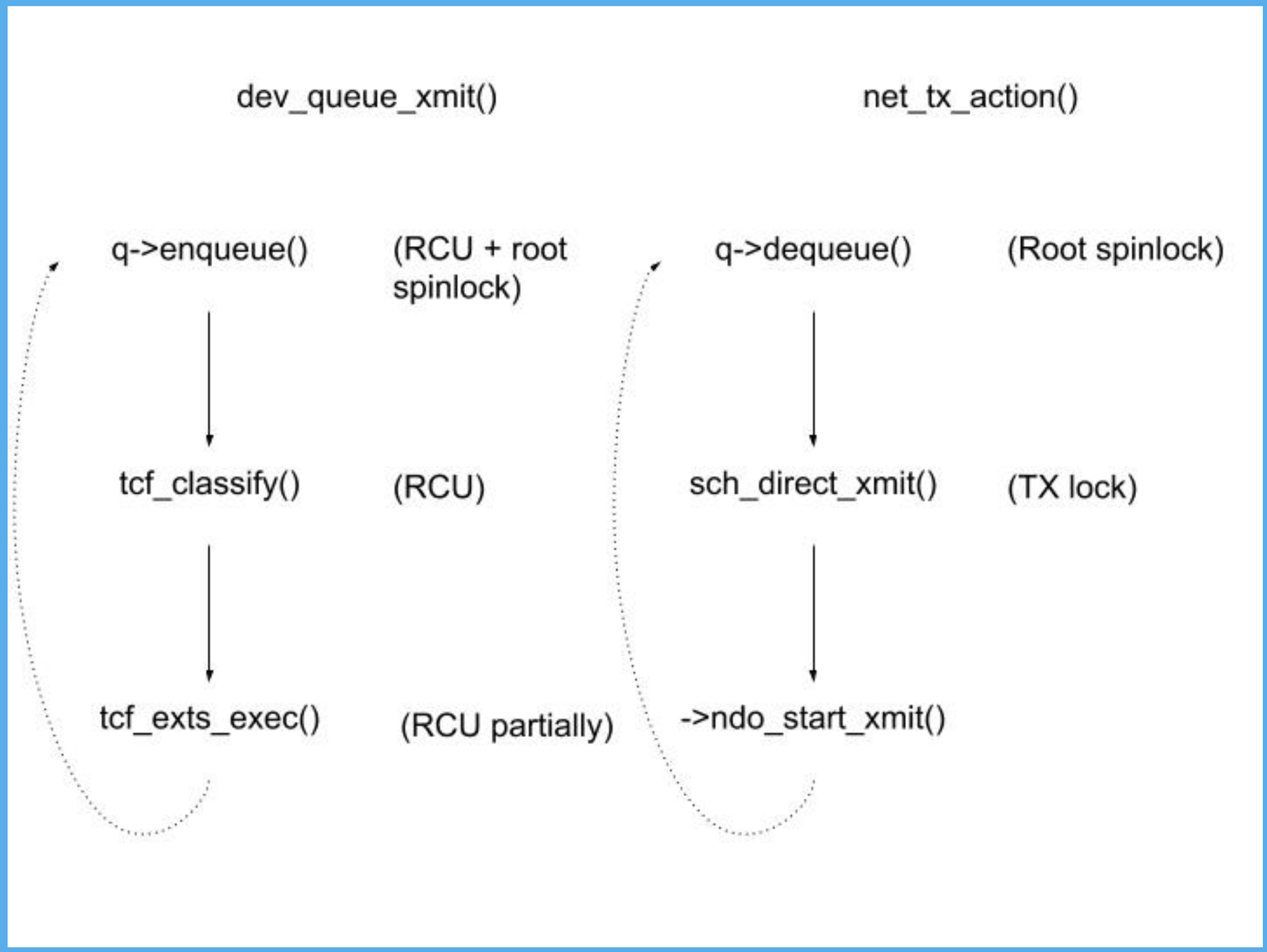


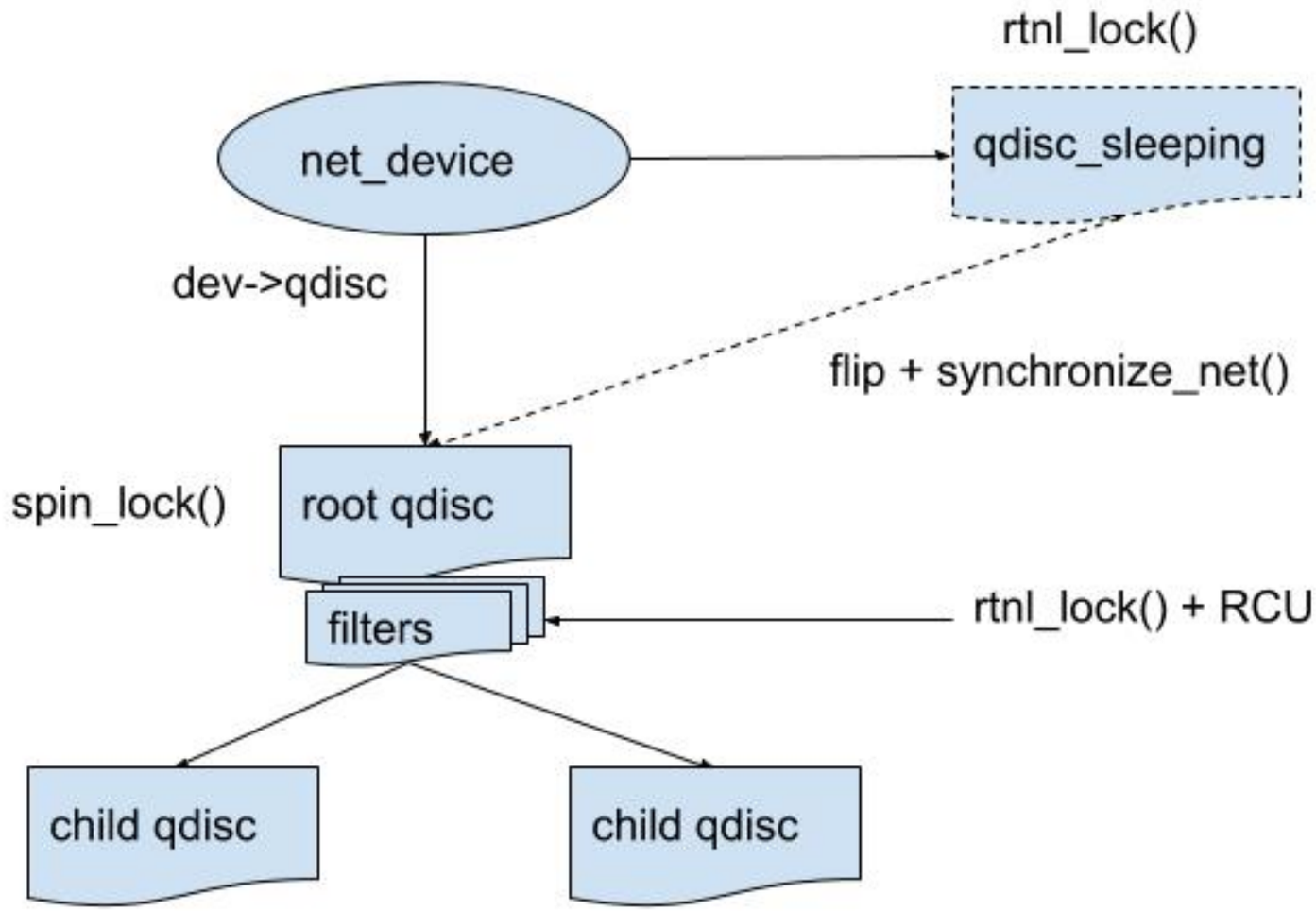
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# Current locking

- RTNL lock on slow path
- RCU read lock on fast path
- Spinlock to sync enqueue with dequeue
- Lock root qdisc
- Lockless on ingress



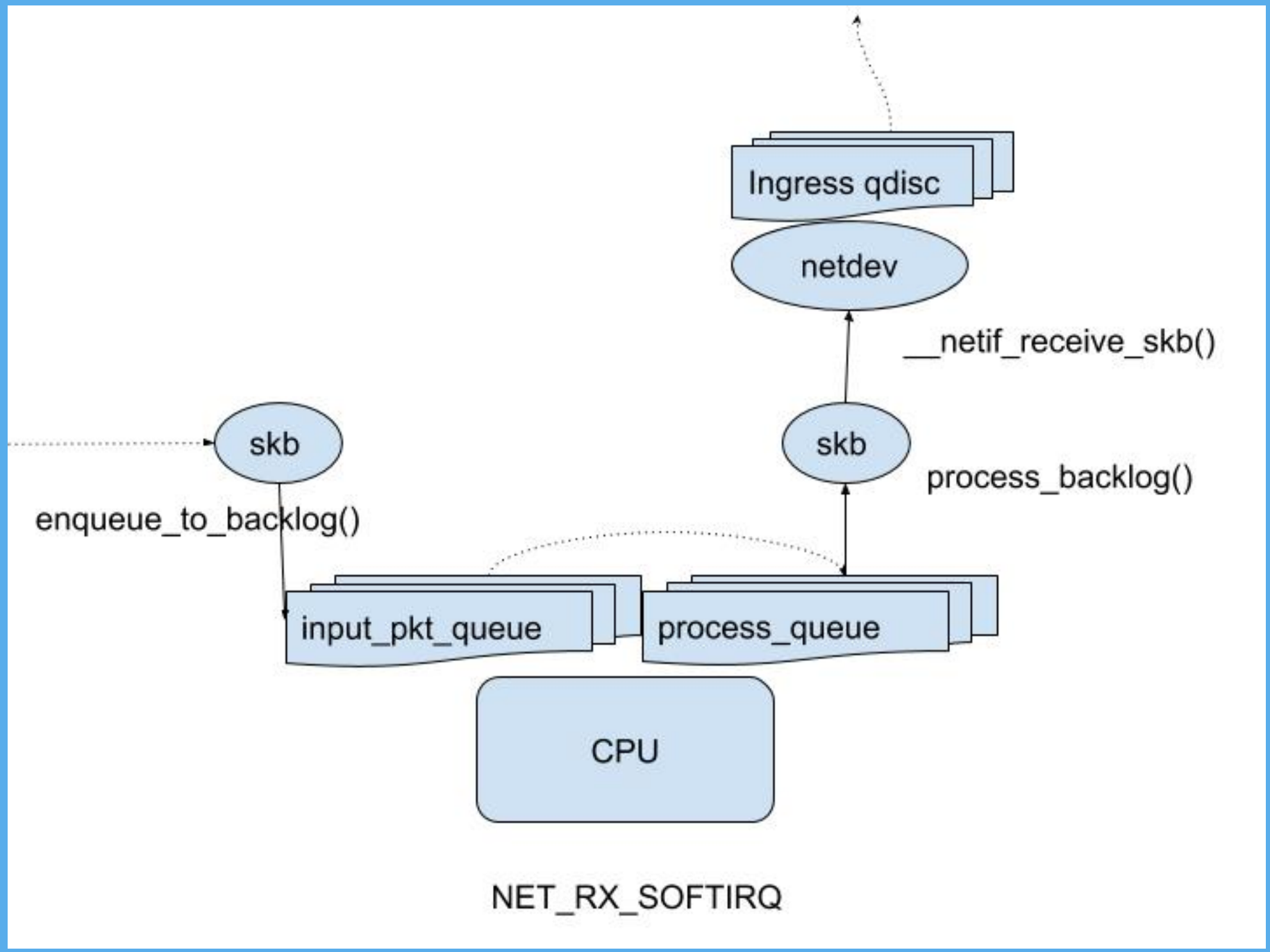

dev_queue_xmit()
net_tx_action()
q->enqueue()
(RCU + root spinlock)
q->dequeue()
(Root spinlock)
tcf_classify()
(RCU)
sch_direct_xmit()
(TX lock)
tcf_exts_exec()
(RCU partially)
->ndo_start_xmit()




Fast path vs. slow path

| | qdisc | filter and embedded action | standalone action |
|---|---|---|---|
| add | dev->qdisc_sleeping | dev->qdisc | idr |
| delete | dev->qdisc_sleeping | dev->qdisc | idr |
| dump | dev->qdisc | dev->qdisc | idr |
| sync with fast path | synchronize_net() | call_rcu() | Filter's call_rcu() |




Ingress qdisc
netdev
__netif_receive_skb()
skb
process_backlog()
skb
enqueue_to_backlog()
input_pkt_queue
process_queue
CPU
NET_RX_SOFTIRQ

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# RCU completeness

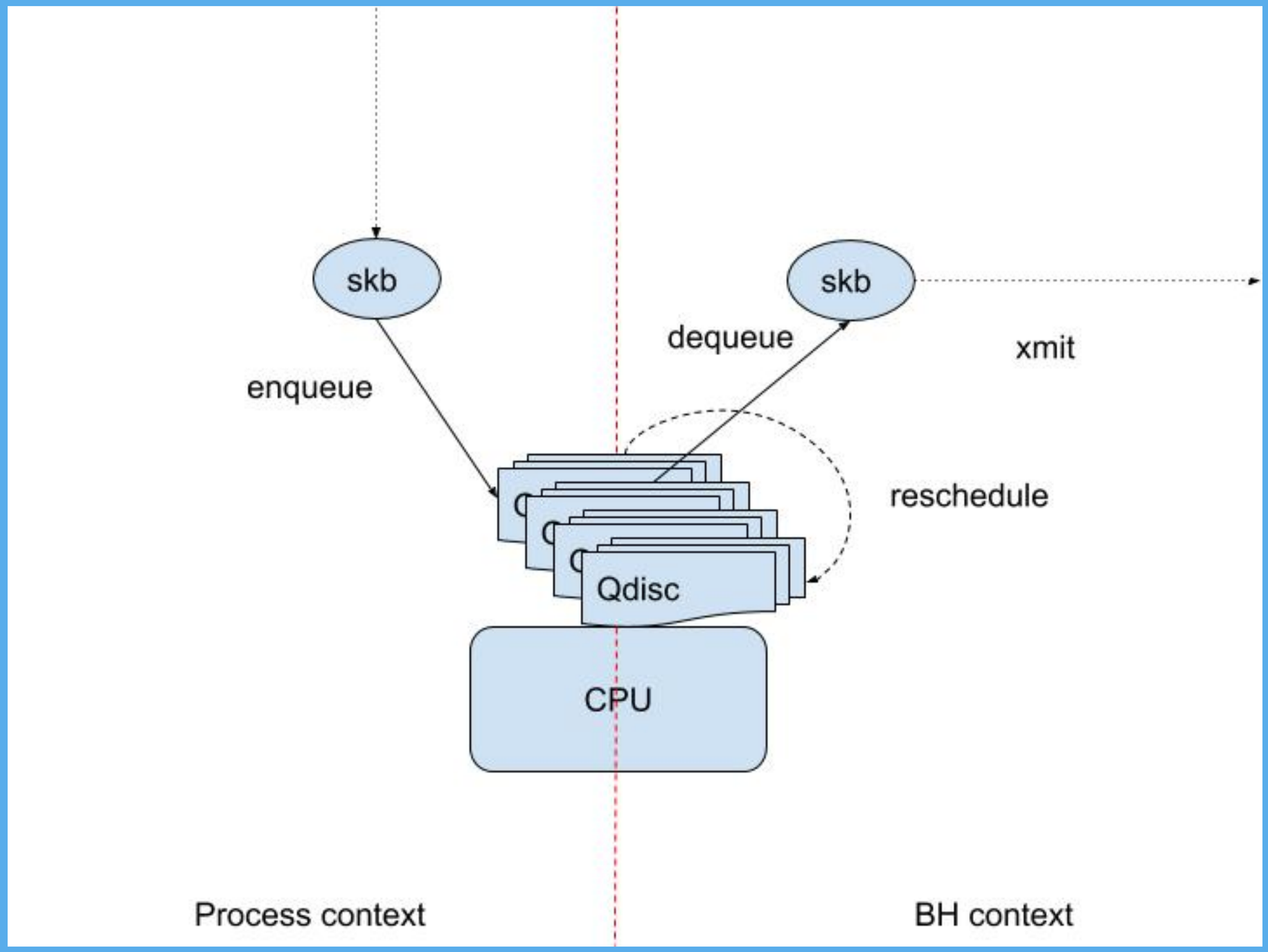
- Qdisc layer still has spinlocks on fast path
- Filter layer is already RCU-complete
- Action layer, copy is still missing
- Completely get rid of TC action spinlocks



# Lockless Qdisc?

- It is a ring buffer!
- No tree lock, but consumer and producer lock
- Decouple enqueue and dequeue
- FIFO is easy, it is class-less and filter-less




skb
skb
dequeue
xmit
enqueue
reschedule
Qdisc
CPU
Process context
BH context

# Really lockless?

- skbs enqueued and dequeued on fast path
- Qdisc can be classful and hierarchical
- BH context vs process context
- Different internal data structures



# Lockless list?

- NULL terminated single linked list, LIFO
- No counting or size limit
- Still have to lock del_first()
- Totally batch enqueue and dequeue?



# Locking ideas

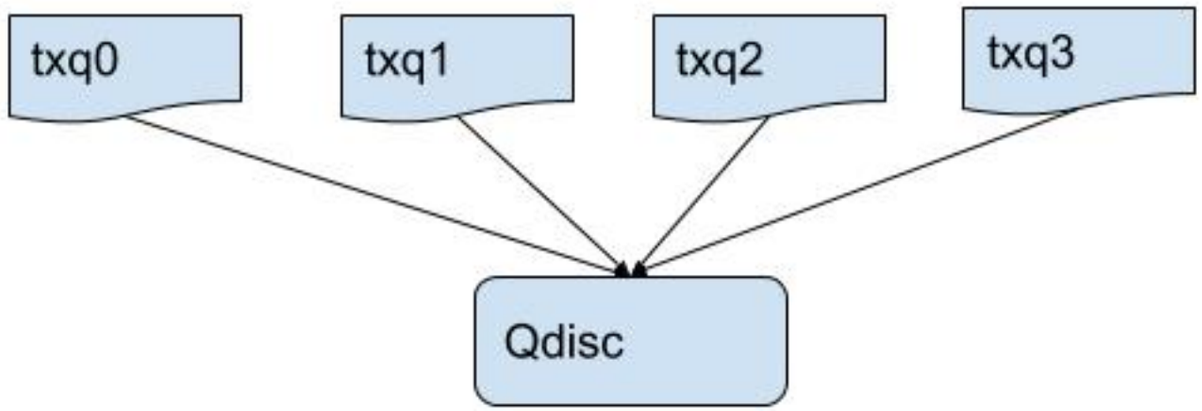
- dev->qdisc_sleeping can be eliminated?
- Break down RTNL to per device locking?
- Break down tree lock to each impl./layer?
- Break down tree lock down to each tx queue?

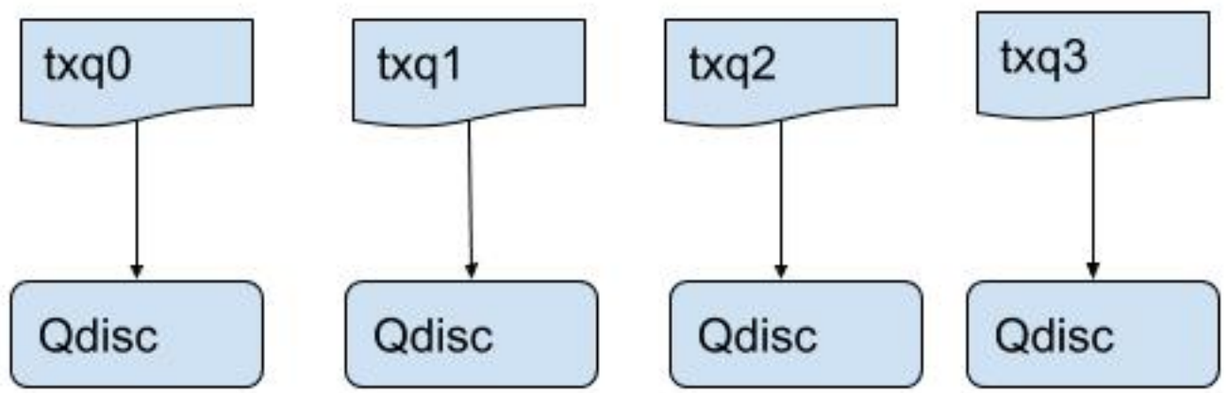


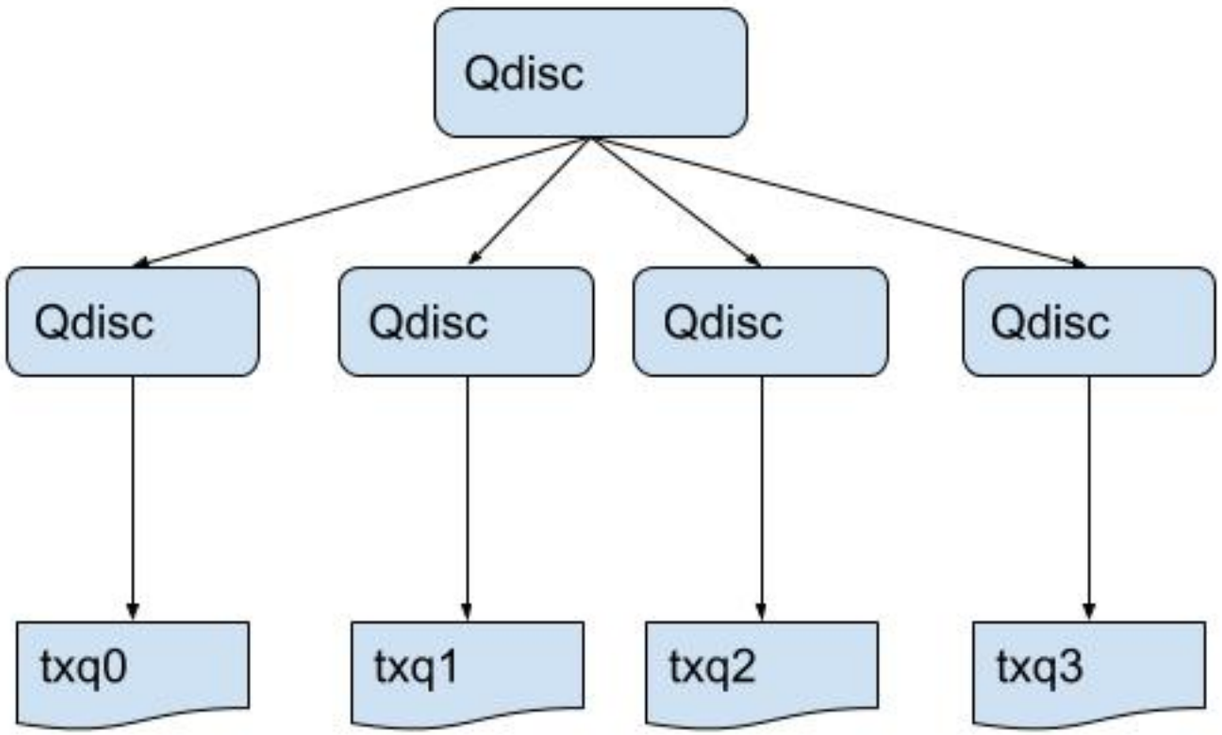
# Tx queue mapping

- A qdisc is shared by all TX queues and CPU's
- Queue selection is done before enqueue
- Except mq qdisc
- Moving TX queue under Qdisc control?
- XPS




txq0
txq1
txq2
txq3
Qdisc


txq0
txq1
txq2
txq3
Qdisc
Qdisc
Qdisc
Qdisc


Qdisc
Qdisc
Qdisc
Qdisc
Qdisc
txq0
txq1
txq2
txq3

# Thank You!



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